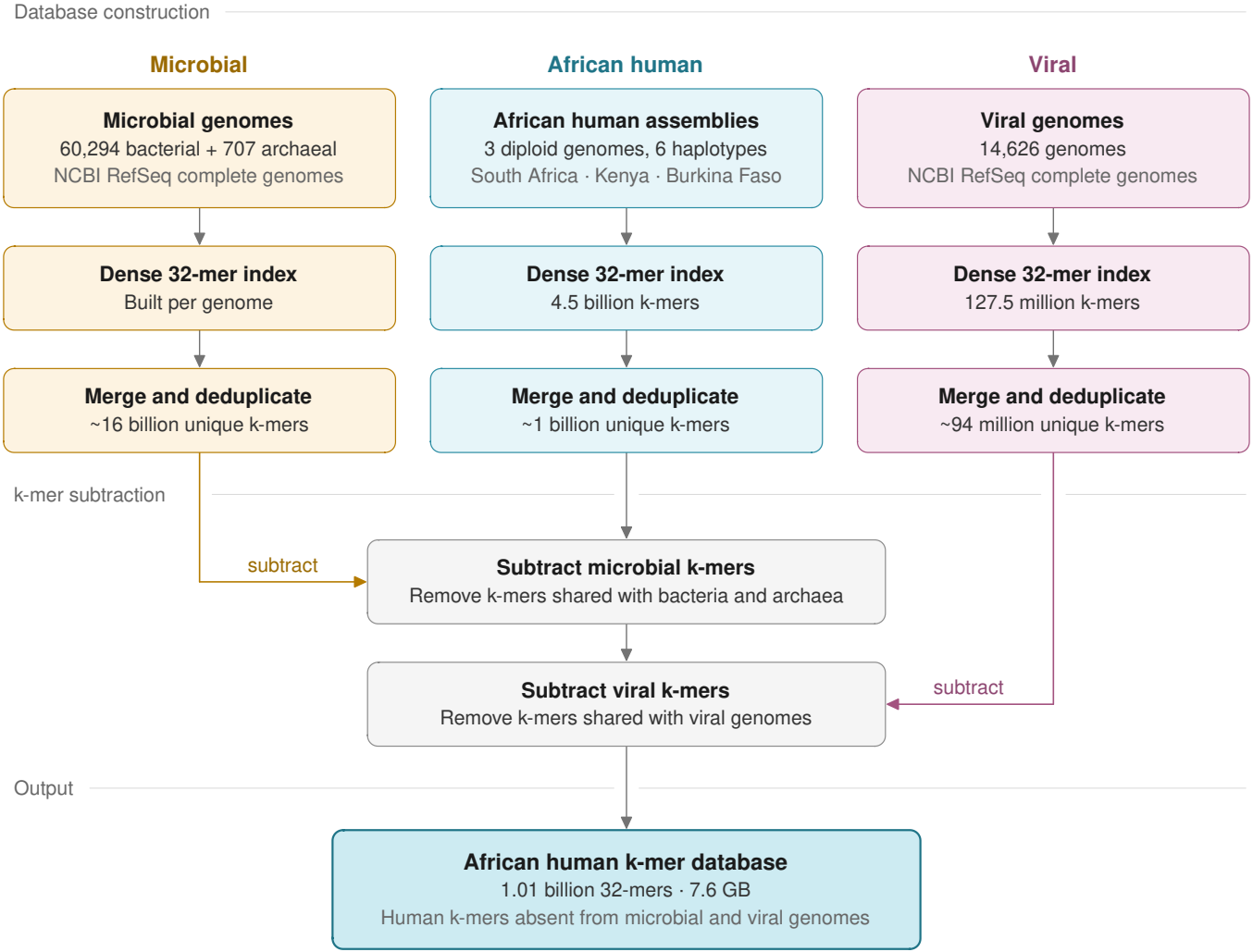
Database construction
Microbial
African human
Viral
Microbial genomes
60,294 bacterial + 707 archaeal
NCBI RefSeq complete genomes
African human assemblies
3 diploid genomes, 6 haplotypes
South Africa · Kenya · Burkina Faso
Viral genomes
14,626 genomes
NCBI RefSeq complete genomes
Dense 32-mer index
Built per genome
Dense 32-mer index
4.5 billion k-mers
Dense 32-mer index
127.5 million k-mers
Merge and deduplicate
~16 billion unique k-mers
Merge and deduplicate
~1 billion unique k-mers
Merge and deduplicate
~94 million unique k-mers
k-mer subtraction
subtract
Subtract microbial k-mers
Remove k-mers shared with bacteria and archaea
Subtract viral k-mers
Remove k-mers shared with viral genomes
subtract
Output
African human k-mer database
1.01 billion 32-mers · 7.6 GB
Human k-mers absent from microbial and viral genomes